

Project Title: Snapdragon Edge-Compute Ground Station for Autonomous UAVs

1. The Problem

Small drones and UAVs have tight weight and battery constraints, making it impossible to carry heavy processors for advanced AI vision tasks (like obstacle avoidance or target tracking). Sending live video feeds to a cloud server introduces high latency, requires a strong internet connection, and drains the battery rapidly. For remote field operations, cloud dependency is a major point of failure.

2. The Solution

Our solution uses a Snapdragon-powered PC as a ground-based "AI Brain." Instead of the drone doing the heavy lifting, it transmits a lightweight camera feed to the laptop. The laptop runs a high-performance object detection model completely offline using its NPU. Once a decision is made, the laptop sends simple, lightweight movement commands (e.g., "stop", "turn left") back to the drone's flight controller via a telemetry radio module.

3. Technical Implementation

- **AI Model:** We utilize a pre-trained computer vision model (like YOLOv8 or YOLOv11) downloaded directly from the Qualcomm AI Hub.
- **Hardware Acceleration:** The model is compiled and optimized to run natively on the Snapdragon Hexagon NPU using ONNX Runtime. This ensures maximum inference speed while keeping the CPU/GPU free.
- **The Bridge:** A Python application processes the NPU's bounding box outputs and translates them into serial commands for an attached radio transmitter, bridging the PC's AI with physical hardware.

4. Impact and Innovation

By offloading AI vision to the Snapdragon NPU, the drone remains lightweight and power-efficient. The system operates with zero cloud connectivity, ensuring total data privacy and ultra-low latency. This project perfectly demonstrates how Windows on ARM can be used as a powerful edge-computing hub for real-world robotics.